Patricia Baldrich González

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RNA biologist investigating the molecular mechanisms that determine RNA processing, trafficking, modification, stability, and function across cellular boundaries. Principal Investigator of an independent, federally funded research program in extracellular RNA biology that integrates molecular genetics, RNA biochemistry, genomics, epitranscriptomics, and direct RNA sequencing to define how RNAs are selected, processed, exported, stabilized, and function outside cells.

RESEARCH EXPERTISE

RNA processing and stability • extracellular RNA biology • small RNAs and ARGONAUTE biology • RNA modifications and epitranscriptomics • direct RNA sequencing • RNA trafficking and secretion • RNA-mediated inter-organismal communication • molecular genetics and genomics

CURRENT ROLE

Director of Strategic Initiatives and Research Funding
Genome Center, University of California, Davis (CA, USA). (2024-present)

- Lead an independent research program defining the molecular mechanisms that govern **extracellular RNA processing, export, modification, stability, and inter-organismal signaling**.
- Serve as PI on major NSF-funded projects integrating RNA biology, molecular genetics, RNA biochemistry, genomics, epitranscriptomics, and microbial physiology.
- Develop experimental and computational approaches to investigate extracellular RNA populations, including small-RNA sequencing, direct RNA sequencing, RNA modification profiling, genetic analysis, and biochemical assays.
- **Mentor** PhD students, postdocs, and junior researchers, embedding inclusive practices to foster career development and scientific excellence.
- Contribute to Genome Center strategy by designing focus groups, pilot grant programs, and **community-building initiatives**, while maintaining strong engagement with external partners and funders.

SCIENTIFIC LEADERSHIP AND RESEARCH EXPERIENCE

2022-2024 Senior Research Scientist and Principal Investigator, Donald Danforth Plant Science Center. Research topic: Extracellular RNA and its role as a signaling molecule.

2020-2022 Research Scientist at Meyers Lab, Donald Danforth Plant Science Center. Research topic: Small RNAs involved in cross-kingdom and interspecies communication.

2016-2020 Postdoctoral researcher at Meyers Lab, Donald Danforth Plant Science Center. Research topic: Small RNAs involved in plant-microbe interactions

2014-2016 **European project contract.** Participation in the EURoot project: Enhancing resource Uptake from Roots under stress in cereal crops. Research topic: Rice and maize phenotypes under drought and mycorrhization conditions.

2012 **Visiting Scientist.** "Institute of Plant and Microbial Biology" Academia Sinica, Taipei (Taiwan). Supervisor: Yue-Ie Caroline Hsing**.** Research topic: Bioinformatics analysis of small RNAs and PARE libraries.

2009 **Undergraduate laboratory training**. During my bachelor's, I completed laboratory training under the direction of Dr. J.P Galaud in "Laboratoire de Recherche en Sciences Végétales" at CNRS Toulouse. Research topic: Functional analysis of four Calmodulin-like proteins (CML) of Arabidopsis

ACADEMIC BACKGROUND

2010-2015 Ph.D. in Plant Molecular Genetics, Centre for Research in Agricultural Genomics (Barcelona, Spain). Under the direction of Dr. Blanca San Segundo. Program: Plant Biology and Biotechnology. Research topic: Regulation of miRNAs during biotic stress in rice and Arabidopsis.

2009-2010 Master's degree in Molecular Biotechnology, University of Barcelona, Spain. Research topics: miRNA function in disease resistance in rice; characterization of calcium-dependent protein kinase 4 (CPK4) under abiotic stress.

2004-2009 Biosciences degree in Biological Sciences, University of Barcelona, Spain.

2004 Diplôme du Baccalauréat Général, série S (Sciences), French pre-college entry exam

FUNDED RESEARCH

2023-2028 **PI**, NSF – Dark Dimensions of the RNA Regulome (D2R2). $1.07 million. Title: The Role of Extracellular RNA in Intercellular and Interkingdom Communication

Explores the evolution, mechanisms, and functions of extracellular RNA in intercellular and interkingdom communication across humans, insects, plants, and microbes.

2022-2027 **PI**, NSF – Plant Genome Research Program**.** $1.2 million. Title: Extracellular RNA Produced By Plants: What, Where, How, Who, and Why?

Defines the cellular origins, composition, processing, export, persistence, and molecular functions of extracellular RNA using plants as a genetically tractable system.

2024 **PI**, Danforth Center Enterprise Rent-A-Car Institute. $100,000. Title: Developing a High-Throughput Method for Collecting and Characterizing Root Exudates

Examines how root exudates mediate plant-environment interactions, focusing on the molecular signals that influence microbial communities, nutrient cycling, and stress resilience.

2020 **Co-PI**, Proteomics Center Mini-Grant, University of Missouri – Columbia. $3,472.

2018 **Co-PI**, Private Donation from an Anonymous Donor. Amount: $70,000. Title: The Billion-Bushel Flower

PUBLICATIONS (* denotes corresponding author**)**

35. Ali Ahmed, Keri Cavanaugh, **Patricia Baldrich**, Ioannis Stergiopoulos, Richard Michelmore. Reduction of Aflatoxin Contamination in Maize Kernels Mediated by Host-Induced Gene Silencing (HIGS). Accepted in PNAS

34. Carolina E. Armijos, Richard O' Connell, Blake C. Meyers, **Patricia Baldrich***. (2026) Phase-specific small RNAs involved in the Arabidopsis thaliana – Colletotrichum higginsianum pathosystems. Plant Direct.
33. Marcia Puig-Lluch, *et al.,* **Patricia Baldrich***. (2025) Making conferences in the plant sciences more inclusive through community recommendations. eLife.
32. Ryan DelPercio, Madison McGregor, Stewart Morley, Nazhin Nikaeen, Blake Meyers, **Patricia Baldrich***. (2025) Transcriptional dynamics of nitrogen fixation and senescence in soybean nodules: A dual perspective on host and *Bradyrhizobium* regulation. MPMI.
31. Rachel Jouni, Caroline Henry, Sébastien Bélanger, **Patricia Baldrich**, Blake C. Meyers. (2025) Distinct classes of 21 and 24-nt phasiRNAs suggest diverse mechanisms of biogenesis and function in rice anther development. The Plant Genome.
30. **Patricia Baldrich***. (2025) Pseudouridine is the hidden language of plant RNA translation. Nature Plants. DOI: 10.1038/s41477-025-01926-w
29. Lucía Borniego, Meenu SinglaRastogi, **Patricia Baldrich**, Megha Hastantram Sampangi-Ramaiah, Hana Zand Karimi, Madison McGregor, Blake C. Meyers, Roger W. Innes. (2025) Diverse plant RNAs coat Arabidopsis leaves and are distinct from apoplastic RNAs. PNAS. DOI: 10.1073/pnas.2409090121
28. **Patricia Baldrich***. My Journey: Embracing Cross-Cultural Experiences. Interactions Journal (2024)
27. Francois Vasseur, **Patricia Baldrich**, Tamara Jiménez-Góngora, Luis Villar-Martin, Detlef Weigel, Ignacio Rubio-Somoza. (2024) miR472 Deficiency Enhances Arabidopsis thaliana Defense Without Reducing Seed Production. Molecular Plant-Microbe Interactions. DOI: 10.1094/MPMI-02-24-0011-R
26. Aleksander Blagojevic, **Patricia Baldrich**, Marlene Schiaffini, Esther Lechner, Nicolas Baumberger, Phillippe Hammann, Taline Elmayan, Damien Garcia, Hervet Vaucheret, Blake C. Meyers, Pascal Genschik. (2024). Heat stress promotes Arabidopsis AGO1 phase separation and association with stress granule components. iScience, doi:10.1016/j.isci.2024.109151
25. Thales Henrique Cherubino Ribeiro, **Patricia Baldrich**, Raphael Ricon de Oliveira, Christiane Noronha Fernandes-Brum, Sandra Marisa Mathioni, Thaís Cunha de Sousa Cardoso, Matheus de Souza Gomes, Laurence Rodrigues do Amaral, Kellen Kauanne Pimenta de Oliveira, Gabriel Lasmar Dos Reis, Blake C. Meyers, Antonio Chalfun-Junior. (2024) The floral development of the allotetraploid Coffea arabica L. correlates with a small RNA dynamic reprogramming. Plant Journal, doi: 10.1111/tpj.16713.
24. Nicole Szeluga, **Patricia Baldrich,** Ryan DelPercio, Blake C. Meyers, Margaret H. Frank (2023). Introduction of barnase/barstar in soybean produces a rescuable male sterility system for hybrid breeding. Plant Biotechnology Journal. doi: 10.1111/pbi.14155
23. **Patricia Baldrich*,** Alexander Liu, Blake C Meyers, Vincent N Fondong. (2022) An atlas of small RNAs from potato. Plant Direct. DOI: 10.1002/pld3.466
22. **Patricia Baldrich**, Sébastien Bélanger, Shuyao Kong, Suresh Pokhrel, Saleh Tamim, Chong Teng, Courtney Schiebout, Sai Guna Ranjan Gurazada, Pallavi Gupta, Parth Patel, Hamid Razifard, Mayumi Nakano, Ayush Dusia, Blake C Meyers, Margaret H Frank. (2022). The evolutionary history of small RNAs in Solanaceae. Plant Physiology, kiac089, https://doi.org/10.1093/plphys/kiac089.
21. Hana Zand Karimi, **Patricia Baldrich**, Brian D. Rutter, Lucia Borniego, Zajt KK, Meyers BC, Innes RW. (2022) Arabidopsis apoplastic fluid contains sRNA- and circular RNA-protein complexes that are located outside extracellular vesicles. The Plant Cell. Apr 26;34(5):1863-1881. doi: 10.1093/plcell/koac043.
20. Thibaut Hacquard, Marion Clavel, **Patricia Baldrich**, Esther Lechner, Imma Pérez-Salamó, Mikhail Schepetilnikov, Benoît Derrien, Marieke Dubois, Philippe Hammann, Lauriane Kuhn, Danaé Brun, Nathalie Bouteiller, Nicolas Baumberger, Hervé Vaucheret, Blake C Meyers,

Pascal Genschik. (2022). The Arabidopsis F-box protein FBW2 targets AGO1 for degradation to prevent spurious loading of illegitimate small RNA. Cell Reports 2022 Apr 12;39(2):110671. doi: 10.1016/j.celrep.2022.110671.

19. **Patricia Baldrich**, Saleh Tamim, Sandra Mathioni, Blake Meyers. (2021). Ligation bias is a major contributor to nonstoichiometric abundances of secondary siRNAs and impacts analyses of microRNAs. (BioRxiv)
18. Wang L, Zhu T, Rodriguez JC, Deal KR, Dubcovsky J, McGuire PE, Lux T, Spannagl M, Mayer KFX, **Patricia Baldrich**, Blake C. Meyers, Huo N, Gu YQ, Zhou H, Devos KM, Bennetzen JL, Unver T, Budak H, Gulick PJ, Galiba G, Kalapos B, Nelson DR, Li P, You FM, Luo MC, Dvorak J. (2021). Aegilops tauschii genome assembly Aet v5.0 features greater sequence contiguity and improved annotation. Genes, Genomes,Genetics G3. Dec; 11(12): jkab325.
17. **Patricia Baldrich,** Tim Chaya, Jeffrey L. Caplan, Blake C. Meyers. (2021) Genome Assembly and Transcriptome of Colletotrichum sublineola CsGL1, a New Resource to Study Anthracnose Disease in Sorghum. Molecular Plant Microbe Interactions. Oct;34(10):1209-1211.
16. Sonia Campo, Francesc Sánchez-Sanuy, Rosany Camargo-Ramírez, Jorge Gómez-Ariza, **Patricia Baldrich**, Lidia Campos-Soriano, Mauricio Soto-Suárez, Blanca San Segundo (2020). A novel Transposable element-derived microRNA participates in plant immunity to rice blast disease. Plant Biotechnology Journal.
15. Vasti T. Juárez-González, Brenda A. López-Ruiz, **Patricia Baldrich**, Eduardo Luján-Soto, Blake C. Meyers, and Tzvetanka D. Dinkova. (2019) The explant developmental stage profoundly impacts small RNA-mediated regulation at the dedifferentiation step of maize somatic embryogenesis. Scientific Reports, 10;9
14. **Patricia Baldrich** and Blake C. Meyers. (2019) Bacteria send messages to colonize plant roots. Science, 365 (6456)
13. Adrien Trolet, **Patricia Baldrich**, Marie-Claire Criqui, Marieke Dubois, Marion Clavel, Blake C. Meyers and Pascal Genschick. (2019) Cell Cycle-Dependent Regulation and Function of ARGONAUTE1 in Plants. The Plant Cell, 31 (8).
12. **Patricia Baldrich**, Brian D. Rutter, Hanna Zand Karimi, Ram Podicheti, Blake C. Meyers, Roger W. Innes. (2019) Plant Extracellular Vesicles Contain Diverse Small RNA Species and Are Enriched in 10 to 17 Nucleotide "Tiny" RNAs. The Plant Cell, 31 (2).
11. Kun Huang, **Patricia Baldrich**, Blake C. Meyers, Jeffrey L. Caplan. (2019) sRNA-FISH: versatile fluorescent in situ detection of small RNAs in plants. The Plant Journal
10. Raquel Salvador-Guirao, **Patricia Baldrich**, Yue-Ie Caroline Hsing and Blanca San Segundo. (2018). DCL1 overexpression increases rice susceptibility to fungal pathogen Magnaporthe oryzae. Annals of Botany, 123.
9. Qili Fei, Yu Yu, Li Liu, Yu Zhang, **Patricia Baldrich**, Qing Dai, Xuemei Chen and Blake C. Meyer (2018). Biogenesis of a 22-nt microRNA in Phaseoleae species by precursor-programmed uridylation. PNAS, 115.
8. **Patricia Baldrich**, Aleksandra Beric and Blake C. Meyers. (2018). Despacito: the slow evolutionary changes in plant microRNAs. Current Opinion in Plant Biology, 42.
7. Raquel Salvador, **Patricia Baldrich**, Ignacio Rubio-Somoza, Detlef Weigel and Blanca San Segundo. (2017) The microRNA miR773 is involved in the Arabidopsis immune response to fungal pathogens. Molecular Plant Microbe Interactions, 31.
6. **Patricia Baldrich**, Mauricio Soto-Suárez, Detlef Weigel, Ignacio Rubio-Somoza and Blanca San Segundo. (2017). The Arabidopsis miR396 mediates pathogen-associated molecular pattern-triggered immune responses against fungal pathogens. Scientific Reports 7.
5. **Patricia Baldrich**, Yue-Ie Caroline Hsing and Blanca San Segundo. (2016) Genome-wide analysis of polycistronic microRNAs in cultivated and wild rice. Genome Biology and Evolution 13;8(4):1104-14. DOI: 10.1093/gbe/evw062.
4. **Patricia Baldrich** and Blanca San Segundo. (2016). MicroRNAs in Rice Innate Immunity. Rice. 9(1):0-6. DOI: 10.1186/s12284-016-0078-5

3. **Patricia Baldrich**, Sonia Campo, Ming-Tsung Wu, Tze-Tze Liu, Yue-Ie Caroline Hsing and Blanca San Segundo. (2015) MicroRNA-mediated regulation of gene expression in the response of rice plants to fungal elicitors. RNA Biology. 12(8):847-63. DOI: 10.1080/15476286.2015.1050577
2. **Patricia Baldrich**, Kakar Klementina, Siré Christelle, Moreno Ana Beatriz, M. García-Chapa, J.J. López-Moya, J.L. Riechmann and Blanca San Segundo. (2014) Small RNA profiling reveals regulation of Arabidopsis miR168 and heterochromatic siRNA415 in response to fungal elicitors. BMC Genomics. 15:1083-1099.
1. Sonia Campo, **Patricia Baldrich**, Joaquima Messeguer, Eric Lalanne, María Coca and Blanca San Segundo. (2014) Overexpression of a calcium-dependent protein kinase confers salt and drought tolerance in rice by preventing membrane lipid peroxidation. Plant Physiology. 165 (2):688-704.

MANUSCRIPTS SUBMITTED AND IN PREPARATION

1. Tanner Smith, Caroline Henry, M. Lucía Borniego, Meenu Singla-Rastogi, Megha H. Sampangi-Ramaiah, Roger W. Innes, Blake C. Meyers, **Patricia Baldrich*.** Conserved and species-specific signatures define plant extracellular small RNA populations. In preparation.
2. Tanner Smith, Ang-Yu Liu, M. Lucía Borniego, Meenu Singla-Rastogi, Roger W. Innes, **Patricia Baldrich*.** Differential Modification and 3' Cleavage affects tRNA Extracellular Fate. In preparation.
3. Caroline Henry, Leonardo Chavez, Michael Wei, Joseph Duenwald, Katia Gutierrez, Katherine M Murphy, Russell Williams, **Patricia Baldrich*.** In Situ Rhizosampling: A Non-Invasive Approach to Capture Root Exudates and Rhizosphere Microbiomes in Soil-Grown Maize. Submitted.
4. M. Lucía Borniego, Meenu Singla-Rastogi, Megha H. Sampangi-Ramaiah, Ang-Yu Liu, Giovanni Gonzalez-Gutierrez, Sarah J. Cox-Vázquez, Chi-Tam Vo, Akihito Fukudome, Ricardo Javier Vázquez, Blake C. Meyers, **Patricia Baldrich**, Roger W. Innes. Extracellular Vacuole-derived bodies (EVacs) mediate RNA secretion in plants. Submitted
5. Summer Blanco, Nicole Cabrera Salazar, Marcia Puig-Lluch, Claudia Garnica-Díaz, **Patricia Baldrich**, Brandi Sigmon, Mary Williams**.** Social Identity Matters in Mentoring (SIMI Mentoring): A Free, Online Training for Culturally Responsive Mentoring in the Plant Sciences. Submitted.
6. Mazen Alazem, Jessi Kreder, **Patricia Baldrich,** Samantha P. Nuzzi, Tessa M Burch-Smith. Reprogramming of auxin and brassinosteroid signaling is an early part of the homeostatic response to a viral movement protein. Submitted and in BioRxiv https://www.biorxiv.org/content/10.64898/2026.02.14.705926v1
7. Emilie Vantard, **Patricia Baldrich**, Marlene Schiaffini, Etienne Reboul, Philippe Hammann, Esther Lechner, Marion Clavel, Christophe Ritzenthaler, Marie-Claire Criqui, Blake C. Meyers, Pascal Genschik. DRB2 is required for efficient and accurate DCL4 processing in plant antiviral immunity. In preparation.

INVITED TALKS AND SEMINARS

2027 Upcoming **Plenary Speaker**, ASPB Annual Meeting (Albuquerque, NM, USA)
2027 Upcoming **Keynote Speaker**, Southern Section ASPB Annual Meeting (Atlanta, GA, USA)
2027 Upcoming Invited Speaker, Plant and Animal Genome Conference, Soybean Genomics Session (CA, USA)
2027 Upcoming Invited Speaker, Integrated Genomics Graduate Group at UC Davis (CA, USA)
2027 Upcoming Invited Speaker, Plant Biology Graduate Group at UC Davis (CA, USA)
2026 BayLeaf Early Career Symposium at UC Davis (California, USA)
2026 Center for Research in Agricultural Genomics (Barcelona, Spain)
2026 Plant Resilience Institute at Michigan State University (Michigan, USA)
2026 Boyce Thompson Institute at Cornell University (New York, USA)

2025 Plant Stress Resilience, Utrecht University (Utrecht, Netherlands)
2025 Plant Genomics Focus Group, UC Davis (California, USA)
2025 Host Microbe Interactions Focus Group, UC Davis (California, USA)
2025 Plant and Animal Genome Conference (PAG – California, USA)
2024 California State University, Northridge (California, USA)
2024 University of Maryland, College Park (Maryland, USA)
2024 HudsonAlpha Institute for Biotechnology (Alabama, USA)
2023 University of North Carolina at Chapel Hill (North Carolina, USA)

TEACHING EXPERIENCE

2026 **Co-teacher** in Genetics Graduate Group **- Student Seminar (Winter and Spring)** to improve presentation skills. University of California, Davis.
2025 **Invited lecture** at Utrecht University, Netherlands
2025 **Co-teacher** in Genetics Graduate Group **- Student Seminar (Winter and Spring)** to improve presentation skills. University of California, Davis.
2023 **Invited lecturer on small RNAs** at Washington University in St. Louis (MO, USA).
2019 **Lecturer at the Teachers Workshop in Molecular Biology**, St. Louis Community College at the Bio Research & Development Growth (BRDG) Park (MO, USA).
2019 Delivered **three lectures** in "Current Approaches in Plant & Microbial Research", Washington University in St. Louis (MO, USA).

MENTORSHIP - External graduate committees

2025- **PhD Committee member** of Gabriana La – University of California, Davis
2025- **PhD Committee member** of Carolina Armijos – University of North Carolina at Chapel Hill.
2024- **PhD Committee member** of Chance Powell – University of Arkansas at Fayetteville.

MENTORSHIP - Current and institutional trainees

2024-2026 **Actively mentoring** two PhD students, three postdoctoral researchers, and one Junior Specialist
2026 **Nate Whittington** – NSF REU Student, 8-week internship at the Genome Center.
2026 **Maxey Mapp** – NSF REU Student, 8-week internship at the Genome Center.
2025 **Owen Black** – NSF REU Student, 8-week internship at the Genome Center.
2025 **Jason Cartwright** – NSF REU Student, 8-week internship at the Genome Center.
2024 **Emma Kramer** – NSF REU Student, 10-week internship at the Danforth Center.
2024 **Cordell Hamilton** – NSF REU Student, 10-week internship at the Danforth Center.
2020-2025 **Rachel Jouni** – PhD Student at Donald Danforth Plant Science Center (MO, USA)
2020-2021 **Pallavi Gupta** – PhD Student at Donald Danforth Plant Science Center (MO, USA)
2016-2022 **Aleksandra Beric –** PhD Student at Donald Danforth Plant Science Center (MO, USA)
2018 **Courtney Schiebout** – Master's student, full-time summer intern (3 months) at Donald Danforth Plant Science Center (MO, USA)
2017 **Jenny Probst** – NSF REU Student, 12-week internship at Danforth Center. Selected REU representative for the National REU meeting in Washington, DC. Project: "tasiRNA function in *Arabidopsis thaliana* during viral infection"
2015 **Claudia Arasa** – Undergraduate student, 6-month full-time lab experience. Autonomous University of Barcelona (UAB, Spain). Project: "Maize phenotype under drought and mycorrhization conditions"
2014-2016 **Raquel Salvador** – PhD student, Autonomous University of Barcelona (UAB, Spain). Project: "Role of plant microRNAs in disease resistance"

2014 **Emilio García** – Undergraduate student, 6-month full-time lab experience. University of Barcelona (UB, Spain). Project: "Role of miRNAs during biotic stress in *Arabidopsis*"

2013 **Ming Tsung Wu** – Visiting PhD Student from Academia Sinica (Taipei, Taiwan); 2-month training

2011 **Giacomo De Simone** – Master's student, 6-month full-time lab experience in MSC Program. Autonomous University of Barcelona (UAB, Spain). Project: "miRNAs involved in plant-pathogen interactions"

SCIENTIFIC SERVICE

Active leadership in scientific societies, diverse initiatives, and mentoring networks, reflecting my sustained commitment to building a more inclusive and collaborative scientific enterprise.

2024-2027 Faculty Adviser, Junior Members of the **RNA Society**

2023-2027 **Board of Directors and Secretary**, International Society of Molecular Plant-Microbe Interactions (**IS-MPMI**)

2026 Co-Lead Organizer of **the Bay Area Plant Hub** scientific meeting with 500 attendees**.**

2026 **National Science Center, Poland –** Ad hoc Review Member

2026 Moderator of the **Seed Central event**, **discussion with the Dean of the College of Agriculture at UC Davis.**

2026 **National Science Foundation, USA –** Grant Panel Review Member

2026 **California Department of Food and Agriculture –** Grant Panel Review Member

2025 **National Science Foundation, USA –** Ad hoc Review Member

2025 **National Science Foundation, USA –** Grant Panel Review Member

2025 Volunteer at **the Bay Area Plant Hub** scientific meeting with 400 attendees**.**

2024 Moderator of the **Seed Central Discussion** panel at the intersection of plant genomics and private companies.

2024-2025 Member, **Diversity, Equity, and Inclusion Committee**, Genome Center at UC Davis

2024 **National Science Foundation, USA –** Grant Panel Review Member

2024 **German Research Foundation (DFG) –** Ad hoc Review Member

2022 **German Research Foundation (DFG) –** Ad hoc Review Member

2020-2022 Member, **Diversity, Equity, and Inclusion Committee,** IS-MPMI

2020-2023 **Junior member**, **Board of Directors**, IS-MPMI

2020 Founder and organizer, **Sustainability Group**, Donald Danforth Plant Science Center (MO, USA)

2020 Coordinator, "CSTM Lunch with the Speaker" Weekly Seminar Series, Donald Danforth Plant Science Center

2019 Co-organizer, **Wells Fargo Innovation Incubator (IN2)**, Donald Danforth Plant Science Center. Collaborated with Dr. Claire Kinlaw to help early-stage companies develop and de-risk technologies through partnerships with Danforth Center scientists.

2019 Founder and organizer, Plant-Microbe Interactions (PMI) Club, Donald Danforth Plant Science Center. Facilitated monthly meetings to share research findings and techniques among scientists.

2017-2018 Co-chair, Committee for Scientific Mentoring and Training (CSTM), Donald Danforth Plant Science Center. Organized three workshops, a career fair, and several networking events for over 200 participants; participated in faculty meetings and retreats.

2014 Co-founder and organizer, 1st Annual Congress of Young Researchers (ACYR), Center for Research in Agricultural Genomics (Barcelona, Spain). Established a recurring event providing PhD students and postdocs a platform to present their work.

2012-2014 Co-chair and PhD student representative, Center for Research in Agricultural Genomics (Barcelona, Spain). Organized mentoring seminars and workshops; participated in faculty meetings.

OUTREACH AND PUBLIC ENGAGEMENT

2025 **Organizer and participant**, Teach the teacher workshop on traditional and molecular breeding for high school and community college teachers.

2022-2024 **Organizer and participant**, Jackie Joyner-Kersee Community Center Outreach Program (IL, USA). Worked with first- and second-grade classes to lead five interactive science activities, enhancing understanding of the scientific method through hands-on learning.

2019-2020 **Organizer and participant**, St. Louis Language Immersion School Outreach Program (MO, USA). Led four classroom activities for 4th-grade students to promote scientific inquiry and consistent engagement with the scientific method.

2018-2020 **Participant**, "**Lunch with a Scientist**", St. Louis Community College at the Bio Research & Development Growth (BRDG) Park at the Danforth Center (MO, USA).

2017-2020 **Guide, Educational Tours at the Donald Danforth Plant Science Center** (MO, USA). Led tours for students ranging from 4th to 12th grade to inspire curiosity and interest in plant science.

2017 **Participant**, "Meeting Grand Challenges: Big Ideas, *the Billion-Bushel Flower*" Donald Danforth Plant Science Center. Participated in the inaugural event featuring three interdisciplinary teams competing for seed funding to support innovative plant science proposals.

HONORS

2026 University of California Davis, College of Biological Sciences**, Principles of Community Award**

2024 Selected as one of the **25 Inspiring Women in Plant Biology** by the American Society of Plant Biologists' Women in Plant Biology Committee

2022 Participant, **Dark Dimensions of the RNA Regulome (D2R2**) Ideas Lab, organized by the National Science Foundation (NSF)

2020 First Author, The Plant Cell - second most attention-getting article of 2019 (182 mentions) for "*Plant Extracellular Vesicles Contain Diverse Small RNA Species and Are Enriched in 10- to 17-nucleotide 'Tiny' RNAs"*

ADDITIONAL EDUCATION AND TRAINING

2025 **From Strategy to Results: Turn Your Strategic Plan into an Action Plan.** Nonprofit Leadership Center.

2025 **Nonprofit Financial Stewardship**: Introduction to Nonprofit Accounting and Financial Statements by Eric Schwartz (Harvard Kennedy School)

2024 **Managing People and Performance**, UC Davis.

2023 **Effective Practices of Successful Organizations** by Jim Meffert, Principal Consultant/Strategist, Tecker International

2022 **Science Leadership & Management Series**, Chan Zuckerberg Initiative.
Includes: Building Effective Teams, Delegating for Productivity and Skills Development, Principles of Leadership, Interpersonal Communication, and Giving Feedback

2021-2022 Yearlong course: **Fostering Genuine Relationships in Academia Through Mentorship**, ROOT & SHOOT NSF grant (Movement Consulting)